**The due date for submitting this assignment has passed.**
**Due on 2026-03-11, 23:59 IST.**

**You may submit any number of times before the due date. The final submission will be considered for grading.**

You have last submitted on: 2026-03-11, 08:07 IST

**Note: This assignment will be evaluated after the deadline passes. You will get your score 48 hrs after the deadline. Until then the score will be shown as Zero.**

## Multiple Select Questions (MSQ)

1) Which of the following statements are correct? ***1 point***

- [ ] If $A$ and $B$ are two square matrices, then Rank$(AB)$=Rank$(BA)$.
- [x] If $A$ and $B$ are two $2 \times 2$ square matrices, then Rank$(AB) \leq$ Rank$(B)$.
- [x] If $A$ and $B$ are two $2 \times 2$ square matrices, then Rank$(AB) \leq$ Rank$(A)$.
- [x] If $A$ and $B$ are $n \times n$ matrices of rank $n$, then $AB$ has rank $n$.

Yes, the answer is correct.

Score: 1

Accepted Answers:

If $A$ and $B$ are two $2 \times 2$ square matrices, then Rank$(AB) \leq$ Rank$(B)$.
If $A$ and $B$ are two $2 \times 2$ square matrices, then Rank$(AB) \leq$ Rank$(A)$.
If $A$ and $B$ are $n \times n$ matrices of rank $n$, then $AB$ has rank $n$.

2) Match the vector spaces (with the usual scalar multiplication and vector addition as in $M_{3\times3}(\mathrm{R})$ ) in column A with their bases in column B and the dimensions of the vector spaces in column C in Table : M2W4GA1. ***1 point***

| | Vector space (Column A) | | Basis (Column B) | | Dimension of the vector space (Column C) |
|---|---|---|---|---|---|
| a) | $V = \left\{ \begin{bmatrix} x & y & z \\ 0 & z & x \\ y & 0 & 0 \end{bmatrix} \mid x + 2y = z, \text{ and } x, y, z \in \mathbb{R} \right\}$ | i) | $\left\{ \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \right\}$ | 1) | 6 |
| b) | $V = \{A \mid A \in M_{3\times3}(\mathbb{R}), A \text{ is a symmetric matrix } \}$ | ii) | $\left\{ \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 & 2 \\ 0 & 2 & 0 \\ 1 & 0 & 0 \end{bmatrix} \right\}$ | 2) | 6 |
| c) | $V = \{A \mid A \in M_{3\times3}(\mathbb{R}), A \text{ is an upper triangular matrix } \}$ | iii) | $\left\{ \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \right\}$ | 3) | 2 |

Table : M2W4GA1

Choose the correct option.

- [ ] a → i → 2.
- [ ] b → i → 2.
- [x] a → ii → 3.
- [ ] c → iii → 1
- [ ] b → iii → 2.
- [ ] c → i → 1.

Partially Correct.

Score: 0.33

Accepted Answers:

a → ii → 3.

b → iii → 2.
c → i → 1.

3) Consider the vector space $V = \{\begin{bmatrix} a & b \\ b & c \end{bmatrix} \middle| a, b, c \in \mathrm{R}\}$. The subset $S = \{\begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\}$ is a linearly independent subset of $V$ . Which of the following matrices $A$ are vectors in $V$ such that $S \cup \{A\}$ is a basis for $V$ ? ***1 point***

- [ ] $\begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$
- [x] $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$
- [x] $\begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$
- [ ] $\begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}$
- [x] $\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$
- [ ] $\begin{bmatrix} 1 & -1 \\ -1 & 0 \end{bmatrix}$

Yes, the answer is correct.

Score: 1

Accepted Answers:

$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$
$\begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$
$\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$

4) Let $S = \{v_1, v_2, v_3\}$ be a set of three vectors in $\mathrm{R}^4$. In order to determine the linear dependence/independence of $S$, we construct a matrix $A$ whose $i^{th}$ column is given by $v_i$, for ***1 point***

$i \in \{1, 2, 3\}$. Let $R$ denote the reduced row echelon form (RREF) of $A$, and let $r_{ij}$ denote the $(i, j)^{th}$ entry of $R$. Choose the correct statements from the following.

- [ ] $R$ does not have any row comprising entirely of zeros if $S$ is linearly independent.
- [x] If $k$ is the number of rows in $R$ comprising entirely of zeros, then the dimension of Span($S$) is given by $4 - k$.
- [ ] If $r_{11}$ and $r_{23}$ are the pivots, then $\{v_1, v_2\}$ is a basis for Span($S$).
- [x] If $r_{11}$ and $r_{23}$ are the pivots, then $\{v_1, v_3\}$ is a basis for Span($S$).
- [x] rank($A$) = rank($R$), i.e., row operations do not alter the rank of a matrix.

Yes, the answer is correct.

Score: 1

Accepted Answers:

If $k$ is the number of rows in $R$ comprising entirely of zeros, then the dimension of Span($S$) is given by $4 - k$.
If $r_{11}$ and $r_{23}$ are the pivots, then $\{v_1, v_3\}$ is a basis for Span($S$).
rank($A$) = rank($R$), i.e., row operations do not alter the rank of a matrix.

**Numerical Answer Type (NAT):**

5) Find the rank of the matrix $A$, where $A = [a_{ij}]$ is of order $2021 \times 2021$ and $a_{i,j} = \min\{i, j\}$ , $i, j = 1, 2, \ldots 2021$
[**Hint:** First do it for smaller matrices, such as $3 \times 3$ or $4 \times 4$.]

2021

Yes, the answer is correct.

Score: 1

Accepted Answers:

(Type: Numeric) 2021

***1 point***

6) Consider a subspace $W = \{(x_1, x_2, x_3, x_4, x_5) \mid 8x_1 + 10x_5 = 0 \text{ and } 9x_2 + 6x_4 = 0\}$ of $\mathrm{R}^5$. The dimension of $W$ is

3

Yes, the answer is correct.

Score: 1

Accepted Answers:

(Type: Numeric) 3

***1 point***

7) Consider a matrix $A = \begin{bmatrix} 2 & -4 & 1 & 1 \\ 1 & -3 & 1 & -4 \\ 3 & -7 & 2 & 3 \end{bmatrix}$. The rank of the matrix is

3

Yes, the answer is correct.

Score: 1

Accepted Answers:

(Type: Numeric) 3

***1 point***

8) Consider two matrices $A = \begin{bmatrix} 3 & 9 & 8 \\ -7 & -10 & -9 \\ 2 & 6 & -7 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & -7 & 2 \\ 9 & -10 & 6 \\ 8 & -9 & -7 \end{bmatrix}$. Then Rank$(A)$ - Rank $(B)$ is

0

Yes, the answer is correct.

Score: 1

Accepted Answers:

(Type: Numeric) 0

***1 point***

9) Consider the subspace $V = \{(x, y, z) \mid z = 7x + 5y, \text{ where } x, y, z \in \mathrm{R}\}$ of $\mathrm{R}^3$ and span$(\{(p, 0, 1), (0, q, 1)\})$ = $V$. Find the value of $\frac{1}{pq}$.

35

Yes, the answer is correct.

Score: 1

Accepted Answers:

(Type: Numeric) 35

***1 point***

10) Suppose $V$ is a vector space defined as $V = \{A \mid A \in M_{4\times4}(\mathrm{R}),\ A$ is an upper triangular matrix, and the sum of the diagonal entries is zero $\}$. What is the cardinality of a basis of $V$?

9

Yes, the answer is correct.

Score: 1

Accepted Answers:

(Type: Numeric) 9.0

***1 point***

11) $\mathrm{U}$ is the vector space of all upper triangular matrices of order 3. $\mathrm{V}$ is the vector space of all lower triangular matrices of order 3. The dimensions of the vector spaces $\mathrm{U}, \mathrm{V}, \mathrm{U} \cap \mathrm{V}$ are $a, b, c$ respectively. Find $a + b + c$. The usual rules of addition and scalar multiplication apply for all three spaces.

15

Yes, the answer is correct.

Score: 1

Accepted Answers:

(Type: Numeric) 15

***1 point***